PANIMALAR ENGINEERING COLLEGE

An Autonomous Institution, Affiliated to Anna university, Chennai

(JAISAKTHI EDUCATIONAL TRUST)

IAT1-Question Bank

DATA COMMUNICATION AND NETWORK SECURITY

PART A
1.Draw the IEEE Data Format for WLAN 802.5 ?
2.When you send a message from your phone to a friend over the internet, what are the five key components that make this data communication possible?
3.For n devices in a network, what is the number of cable links required for a mesh, ring, bus, and star topology?
4.Compare and contrast Omni-directional waves and unidirectional waves?
5.Show how data link control services manage error detection and flow control in a given network scenario.
6.Compare and contrast byte-stuffing and bit-stuffing. Which technique is used in byte- oriented protocols? Which technique is used in bit-oriented protocols?
7.While you are having a video call, your device sends and receives data at the same time. Which technique is used to send acknowledgment along with data to improve efficiency, and what is its purpose?
8.Imagine you are sending a long document over a network. How does the system know where one part of the message ends and the next begins? What method is used to organize data for accurate transmission and why is it necessary?
9.Demonstrate how to assign IPv4 addresses to different devices in a small office network while avoiding address conflicts.
10.Evaluate the role of default masks in classful addressing and discuss their limitations in modern network design.
11.Distinguish between point-to-point link and broadcast link.
12.Outline the need for switching and how does it help manage the control signals effectively?
13.For n devices in a network, what is the number of cable links required for a mesh, ring, bus, and star topology?

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14.which network performance parameters would you analyse to diagnose the issue, and how would each parameter help in identifying the network's performance problems?
15.Organize the role of error control and how does it ensure the accurate delivery of data despite potential transmission errors?
16.A large video file is being sent over a network that only supports small packet sizes. How does the network ensure the entire file is transmitted successfully without exceeding size limits? What role does fragmentation play in this process?
17.Suppose the following sequence of bits arrives over a link 110101111101011111001011111 0110. Show the resulting frame after any stuffed bits have been removed. Indicates any errors that might have been introduced into the frame
18.How would you assess the roles of nodes and links in this network, and why are both essential for the network's functionality and performance?
19.Find the class of each address a.11000001 10000011 00011011 11111111 b.252.5.15.111
20.Analyze how routers differentiate the incoming unicast, multicast and broadcast IP packets.
21. Draw the IEEE Data Format for WLAN 802.5 ?
22.Define the Masking with example ?2
23. What is ICMP ?
24.Given a network communication scenario, demonstrate how both the transport layer and the data link layer contribute to reliable data transfer. PART B
1.You are assigned to train new interns at a tech company on how data travels across a network. To help them understand real-world communication between two computers over the internet (such as loading a webpage or sending an email), you decide to use the TCP/IP protocol suite as the foundation of your explanation. In this scenario, how would you explain the layers of the TCP/IP protocol suite? Describe how data is transferred

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across each layer, starting from the application on the sender's side down to the physical transmission, and then back up to the application on the receiver's side.
2.At a college, students often access online resources, attend virtual classes, and transfer files across departments. As a network planner, you must choose the best method for transferring data across the network. What are the different types of switching techniques available, and what are their advantages and disadvantages in handling this campus data traffic
3.As the network administrator of a college campus, you are tasked with designing the layout for connecting computer labs, library systems, faculty offices, and administrative blocks. What are the different network topologies you can consider for this setup, and what are the advantages and disadvantages of each in terms of cost, performance, and reliability?
4.Estimate your idea on how guided media differ from unguided media? Briefly explain any three methods used for data communication using guided media and two methods used for data communication using unguided media.
5.How is High-Level Data Link Control (HDLC) classified based on its operational modes and explain the purpose and structure of each frame type in detail.
6.In an office network, multiple computers are connected and need to share files, access the internet, and communicate efficiently. Which devices manage data traffic in such a setup, and how do switches, hubs, and routers handle the data differently?
7.Analyze the efficiency and reliability trade-offs between Stop-and- Wait, Go-Back-N, and Selective Repeat protocols in various network conditions.
8.Construct the comparison between different wireless technologies? Enumerate 802.11 protocol stacks in detail.
9.You are managing the network for a university campus with multiple departments connected through routers. One routing algorithm being used exchanges information only with neighbouring routers to determine the best path. What is this method called, how does it work compared to other approaches like Link State routing, and what limitations might it present in a large or dynamic network

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10.Demonstrate the role and significance of Internet Control Message Protocol version4 (ICMPv4)? How does it support communication and error reporting in IP networks and explain the types of messages used and their role in network diagnostics.
11.You are responsible for setting up connectivity in a university that includes student labs, inter-department communication, and internet access across campuses. What types of computer networks would be suitable for each scenario, and how do they differ in terms of scale, purpose, and communication?
12.Compare and contrast guided media and unguided media?
13.Discuss how the Data Link layer, Network layer, and Transport layer function in ensuring reliable and accurate data communication. How do these layers collaborate to handle addressing, routing, error detection, and end-to-end delivery? Illustrate your explanation with appropriate diagrams.
14.Explain Cyclic Redundancy Check? Suppose we want to transmit the message 11001001 and protect it from errors using the CRC polynomial $x^3 + 1$. Use polynomial long division to determine the message that should be transmitted.
15.How does error control ensure reliable data transmission between nodes and analyze the different techniques used for flow and error control in Data Link Layer.
16.You are setting up a wireless network in a college campus with multiple access points to support students and faculty. To ensure efficient communication and coverage, what architectural components of IEEE 802.11 need to be considered, and how do they work together to manage wireless connectivity?
17.You are working as a network technician for an ISP that uses Point-to-Point Protocol (PPP) to establish direct communication links with customers over dial-up and leased lines. One of the clients is experiencing issues during link establishment, and you suspect the problem lies in the data framing process. In this scenario, how would understanding the PPP frame format help in diagnosing and resolving the issue? Illustrate and explain the structure of a PPP frame with a neat diagram, describing the purpose of each field in ensuring successful data transmission.

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18.How does Bluetooth technology work to establish and manage communication between devices? Identify its working principles and develop a neat sketch to depict the Bluetooth architecture, explaining how it supports short-range wireless communication.
19.You are a network engineer at a large internet service provider (ISP). Recently, users have been experiencing routing issues when trying to access websites hosted in different autonomous systems across the globe. To resolve this, you need to analyze how routing decisions are made between different networks. In this scenario, how does the Border Gateway Protocol (BGP) help manage inter-domain routing across the internet? Examine the functions of BGP and explain how it ensures efficient, loop-free, and policy-based routing between autonomous systems.
20.Develop the roles of IP and ICMPv4 in network communication. Focus on how IP ensures packet delivery across networks, and explain how ICMPv4 supports error reporting and diagnostics in case of delivery failures.
PART C
1.A student is uploading a project file to a cloud server from their laptop over the internet. Which protocols in the TCP/IP suite are involved in this process, and how do they work together to ensure reliable transmission, addressing, and delivery of data?
2.You are part of a network modernization team for a large organization expanding its digital infrastructure globally. As more devices are added to the network, your team begins facing issues like IP address exhaustion, limited scalability, and inefficient routing using IPv4. In this scenario, how does IPv6 address the challenges faced by IPv4 in modern networks? Discuss how IPv6 overcomes limitations such as address scarcity, NAT dependency, and inefficient header structure, and explain the advantages it offers for future-ready networking.
3.Imagine you are sending an email with an attachment from your computer to a friend abroad. The data travels through several stages before reaching the destination. How do the different layers of the ISO-OSI model contribute to this process, and what are the key functions performed at each layer?
4.A message is to be transmitted using Cyclic Redundancy Check (CRC). Summarize the purpose of CRC in error detection and solve the following problem: Given Message: 11010011101100 Generator Polynomial: $x^4 + x + 1$